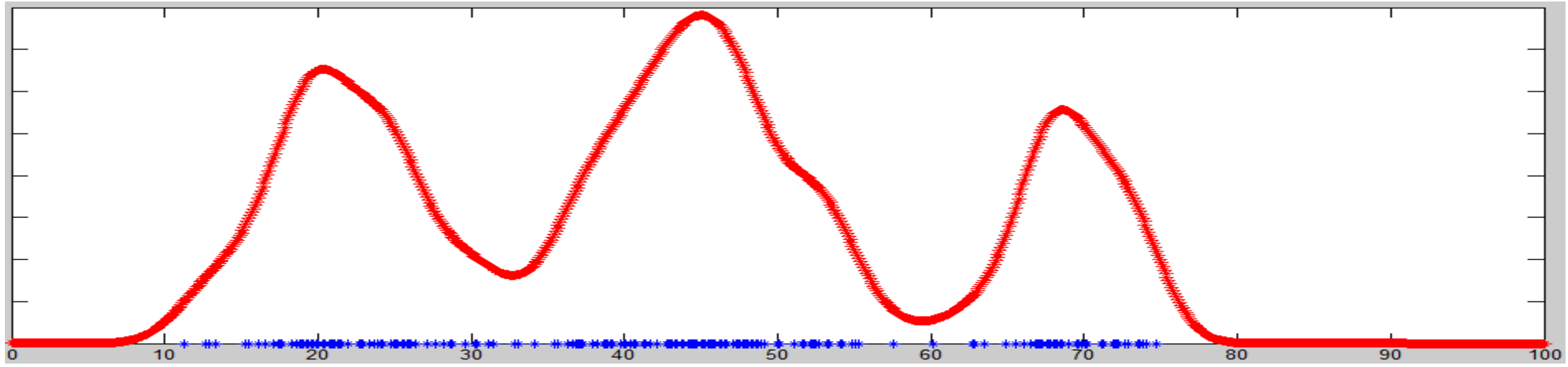
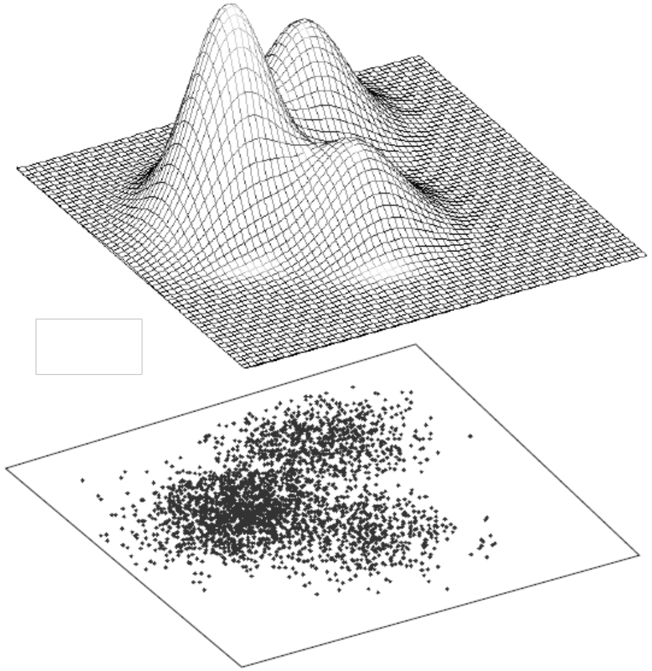
Aside: Kernel Density Estimation

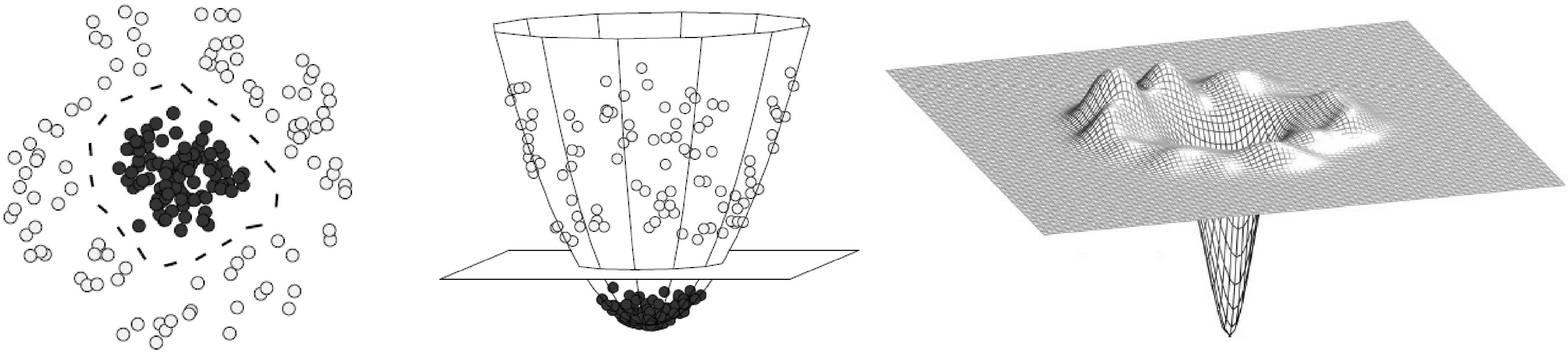
- Can approximate a continuous distribution from a point cloud by a superposition of Gaussian "kernels" centered on all its points.
  - Related: "Mean shift" clustering
  - A mixture of a smaller number of Gaussians can give a much simpler approximation…





Aside^2: Support Vector Machines with Gaussian Kernels

Kernel methods in machine learning often used to “trick” algorithms designed for finding linear answers (e.g., decision boundaries) into finding nonlinear answers (linear answers after mapping to a higher-dimensional space with a “kernel function”).



Expectation Maximization

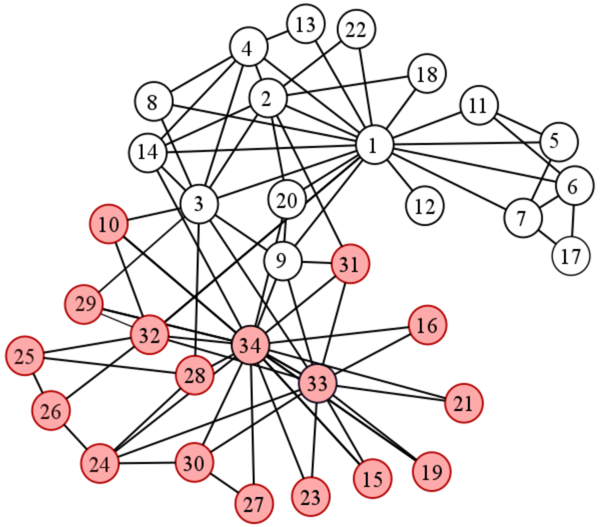
- Common usage: mixture modeling
  - Find parameters of a mixture of k models most likely to have generated our data.

Overall model $= w_1$ **Model 1** $+ w_2$ **Model 2** $+ \ldots + w_k$ **Model k**

- Alternatively update:
  - **Explicit** model parameters and mixture weights
  - **Latent** parameters: e.g., p_{ij} = **Pr**[i^{th} point generated by j^{th} model]
- Used for many other model-fitting tasks (e.g., Hidden Markov Models, Bayesian Networks)

Consensus Among Clusterings

- Generate N clusterings.
- Build “consensus graph” – weight of edge (x, y) is fraction of clusterings in which x and y belong to the same cluster.
- If all weights binary, we’re done.
- Otherwise, use consensus graph to build N clusterings and repeat…



Nonnegative Matrix Factorization

- We often want to “factor” a data matrix to expose an underlying structure given by mixtures of “latent” sources / factors with certain properties – e.g.,
  - Uncorrelatedness (factor analysis / PCA)
  - Independence (independent component analysis)
  - Nonnegativity (nonnegative matrix factorization)
  - Sparsity

Data vector 1 Data vector 2 … Data vector N	=	(mixture weights)	Latent source / factor 1 … Latent source / factor M
A		**X**	**Y**

Nonnegative Matrix Factorization

- Alternating iterative refinement:
  - Hold X fixed, recompute Y to minimize the error in A – XY
  - Hold Y fixed, recompute X to minimize the error in A – XY
- Even though the overall problem is NP-hard, each step above typically set up as a nonnegative least squares problem (much easier).